

Diversity Healthcare

Phase 3

Contributed by:

Project Leader: Anabelle Hsiao

Team Communicator: Muhammad Bilal

Zeynep Sude Haksal

Christian Rogers

Vignesh Veerakumar Sathya

Ho Ching Ashley Wong

Figma Prototype:

Doctor View:

<https://www.figma.com/proto/iZTQXMzdTFQXQ0KwxznBBp/Team-Diversity-Phase-3?node-id=1-15141&t=SLxFX9OPkxF7DsD-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1>

Patient View:

<https://www.figma.com/proto/iZTQXMzdTFQXQ0KwxznBBp/Team-Diversity-Phase-3?node-id=1-14638&t=vw1RliHkHYMZPN7v-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1>

Task Descriptions with Step-by-step Instructions:

Task 1: View/Book/Cancel Appointments:

Patient View(View/Book/Cancel Appointment)

Task Description:

A patient wants to book an appointment. They start by viewing nearby clinics, find the one they prefer and book it at the desired time. Once booked, if the patient wants to edit the information, they may update the time and date. Patients also have the option to cancel and rebook appointments.

Step-by-step Instructions:

1. Homepage → check clinics nearby
2. Check clinics nearby → add appointment
3. Add appointment → Dr Eggward Cullen arrow
4. Dr Eggward Cullen → Accept
5. View Appointments → Dr Eggward Cullen Edit button
6. Edit page → Update button

Task 2: View and Create Patient Records and Prescription Records:

Patient View (Health Records)

Task Description:

A patient wants to view their test results. The health records page contains a scrollable list of test results sorted by most recent, with optional filters and sort settings. There is also a search bar, which allows the user to search by test name. By clicking the arrow button on the right side of the textbox, they can view a result summary and an in-depth view of the result specifics (assuming the test has a result scale/criteria).

Step-by-step Instructions:

1. Homepage → Health Records
2. HealthRecords → Cholesterol Test left arrow
3. Cholesterol Test Page → View results and return to records with top left corner return arrow
4. Health Records → Return home with top left arrow

Doctor View (Health Records Add, View, Update)

Task Description:

A doctor needs to add, update and view health records. The patient page contains a list of health records sorted by date added. Each record has a right arrow which leads to the specific record page containing text fields with record details and a file upload on the left-hand side for DICOM sets (CT, MRI images) or test result sheets. The user can edit the text field and press update to change the record text. Additionally, on the patient record page, there is a, “Add Record” button which takes the user to a new record page where they can enter details of the record.

Step-by-step Instructions:

1. MD-Homepage → John Doe (details button)
2. Brain MRI (Left side third down) → right arrow button
3. See uploaded files, Text fields containing current data (Update Changes record entry)
4. Use the return arrow (top left) to return to the patient record
5. User Record → Add record button (Mid left)
6. Use the return button to return to the user record
7. Use the return button again to return to MD-Home-Page

Patient View (Prescription Orders)***Task Description:***

A patient wants to see prescriptions they need to fill after an appointment. The Prescription Orders Page contains a scrollable list sorted by date ordered of prescriptions added by their doctor. Each Prescription can be accessed by its right arrow, which leads to that prescription order. If the prescription has not been filled the page will contain the drug image, drug details (name, unit price, OHIP coverage) followed by a list of nearby pharmacies to pick up from. The user can click on the details of a listed pharmacy, and it will contain the hours, map location and directions option, and the prescription info, along with a QR code for the pharmacist to scan.

Step-by-step Instructions:

1. Homepage → Prescriptions
2. Atorvastatin (right arrow button on the drug card) → Drug Details
3. Pharmacy 1 Details Button → Location Details and prescription info
4. Back Button → Prescription Info
5. Back Button → Prescriptions
6. Levothyroxine (right arrow button on the drug card) → Drug Details
7. Back Button → Prescriptions
8. Back Button → Home Page

Doctor View (Prescription Orders)***Task Description:***

A doctor needs to view, update and add prescription orders for a given patient. The patient's page contains a list of prescriptions made out for the patient. Each prescription can be accessed using the right arrow button. The order record contains the drug details and the prescription order (text fields). The text fields can be edited, and the update will change their value and return to the user record. The add order button allows the user to order a new prescription. The search bar searches prescriptions by name, and when clicked auto-fills the text field with the drug and unit size. The other text fields can be edited to the prescription specifics, and the order button adds the order to the user record and returns.

Step-by-step Instructions:

1. MD-Homepage → John Doe (details button)
2. Add Order (Center Middle Button) → New Drug Order
3. Use "Order" or back button (upper left to return) → John Doe

4. Atorvastatin (right arrow button) → Update Drug Order
5. Use “Update” or back button (upper left to return) → John Doe
6. Return Button (upper right) to return → MD-Homepage

Task 3: View Clinics Around the Area:

Patient View (View Clinics Nearby)

Task Description:

A patient wants to find nearby clinics and quickly access key details (e.g. distance, opening hours, languages the clinic provides, specialties, etc) before deciding to book an appointment. The “Clinics Nearby” page presents a high-level, scrollable list that is defaultly sorted by distance with concise overview chips for every clinic. There are also toggleable filters (e.g. “Open Now”, “Favourites”, etc) and a search bar. By clicking on the “Details” button on a clinic card, patients are brought to the “Clinic Details” page, where they can drill down on specific details. This top-down approach allows users to scan fast, shortlist clinic options, and only invest time in deep details when they see a clinic they are interested in making an appointment at.

Step-by-step Instructions:

- 1. Homepage → Clinics Nearby Page**
 - At the bottom of the homepage sits the “Check Clinics Nearby” button
- 2. Scan/Scroll the default list**
 - The default list is sorted by distance (closest to the user)
- 3. Search Clinic(optional)**
 - The user can click on the search bar to search for clinics if they know of any. In this prototype, the user searches for clinics with the word “Town”.
- 4. Clinics Nearby Page → Clinic Details Page**
 - The user can click on the “Details” button on a clinic card and they will be brought to the Clinic Details page. In this prototype, the user can click and view details for Clinic Y.
- 5. Read Map and Clinic Details**
 - Each Clinic Details page includes details including the address, opening hours, distance, specialties, languages provided, etc.
 - For “Doctors” and “Next Available Today” in the prototype, the user can scroll horizontally to view these details.
- 6. Proceed to Book an Appointment (optional)**
 - There is a “Book Appointment” button at the bottom of the Clinic Details Page, where users can proceed to make an appointment if they want to.

Evaluation on prototype in the next phase of the project (Phase IV).

In the next phase of our project, our evaluation will focus on verifying whether our design supports the primary tasks identified in earlier phases and whether the interface is usable, learnable, and efficient for both patients and doctors. Specifically, we will evaluate three core end-to-end task flows: viewing, booking, and cancelling appointments; viewing and managing health records and prescription orders; and finding nearby clinics with the option to view details and book appointments. These flows align directly with the task priorities outlined in Phase II, where both patients and doctors emphasized the importance of quick access, minimal navigation steps, and clear presentation of health information.

We will be using heuristic evaluation, cognitive walkthrough, and usability testing methods to evaluate our prototype. These methods will provide design insights by both experts and users, capturing a range of usability issues and validating whether our interface supports the needs of different users.

First, we will conduct a heuristic evaluation to identify usability problems efficiently before user testing. This method is fast, cost-effective, and capable of revealing interface issues such as consistency, visibility, and error prevention. Three evaluators will independently inspect the prototype using Nielsen's ten heuristics, assign severity ratings to each identified issue, and later consolidate their findings into a single list. This method is useful for our prototype because it allows us to catch navigation and feedback inconsistencies among different parts of our application before we involve actual users.

Next, we will perform a cognitive walkthrough to evaluate the learnability of the interface for first-time users. This method will tell us whether users will form the right goal, notice the correct available action, associate the action with the goal, and receive appropriate feedback. For instance, we will examine whether a patient can easily navigate from the homepage to book an appointment, view a lab result, or find a nearby clinic, and whether a doctor can intuitively add or update a prescription record. This method is relevant because our app serves two distinct user types with different mental models and technical familiarity. By walking through these tasks step by step, we can identify moments of confusion, unclear labels, or missing feedback that might hinder learnability, especially for first-time users.

After addressing issues found in the expert evaluations, we will conduct usability testing in a controlled setting with real users to collect both quantitative and qualitative data about how well users can perform tasks using the prototype. Participants will be asked to complete realistic tasks such as booking an appointment, viewing a test result, updating a prescription, and locating an open clinic. During these sessions, participants will verbalize their thoughts as they navigate the prototype, and we will record task completion times, the number of errors, whether users required assistance, and any other information that provides insight into user usability. Each participant will also complete a short survey after their session to assess satisfaction and perceived ease of use. Quantitative measures such as completion rates and error counts will help determine whether users can complete tasks within acceptable time limits and without confusion. Qualitative observations from the think-aloud sessions and debriefs will reveal the reasoning behind user difficulties and suggest design improvements.

Ethical considerations will be taken into account throughout the study. Informed consent will be obtained from all participants, the purpose of the study will be clearly explained, and no personal or sensitive health information will be collected. Participants will be reminded that their involvement is voluntary and that they may withdraw at any time without consequence, following the ethical principles outlined by the UofT Research Ethics Board.